Module 2 — Quiz A

Question bank

Module 2, Quiz A: Definitions & notation

Reading: Module 2 Read 1

> **!** By-hand multiplier 2 (Quiz A & B)
>
> On these quizzes, use 2 as the critical value for Wald proportion CIs and approximate mean CIs unless a item says otherwise. Your endpoints may be slightly different from `prop.test()` ($z^* \approx 1.96$) or `t.test()` (t^* depends on n). That is expected. Synthesis, labs, and Project 2 use R output.

Section A — Confidence intervals & test vs CI (Read 1 §2)

A1. 95% confidence level

Define 95% confidence level in terms of repeated sampling and intervals.

A2. Confidence interval

Define confidence interval in one or two clear sentences.

A3. Test vs CI — research questions

A fair-coin study observes 7 heads in 16 flips.

(a) What question does a hypothesis test of $H_0 : p = 0.5$ answer?

(b) What question does a 95% CI for p answer?

A4. Test vs CI — wording

Which pair best matches test vs confidence interval?

A) *What is* the population proportion? / *Is* the proportion equal to 0.30?

B) *Is* the proportion equal to 0.30? / *What is* a plausible range for the population proportion?

C) Both ask the same question with different notation.

A5. Incorrect CI interpretation

Which is wrong for a single 95% CI you already computed?

A) If we repeated sampling many times, about 95% of such intervals would contain the parameter.

B) There is a 95% probability the parameter lies in this interval.

C) I am 95% confident the true population proportion is between the endpoints.

A6. Test vs CI standard error

One-proportion test uses p_0 in the SE; one-proportion CI uses $\hat{p}$ in the SE. Why?

Section B — Notation: n, N, p, x, $\hat{p}$ (Read 1)

B1. Part-time job survey

Random sample $n = 120$ undergraduates; $x = 46$ work ≥ 10 hrs/week.

Identify parameter, statistic, and compute $\hat{p}$.

B2. Clinic satisfaction

$n = 80$ patients sampled; $x = 52$ very satisfied.

Compute $\hat{p}$ and state what p represents.

B3. n vs N

A campus has $N = 4{,}200$ students; you measure $n = 80$ in an SRS.

What does each letter denote?

B4. p vs $\hat{p}$

In a vaccine study, 88 of 200 patients showed a response.

Identify x, n, $\hat{p}$, and the parameter p.

B5. Parameter

Define parameter in one or two clear sentences (proportion or mean example).

B6. Statistic

Define statistic in one or two clear sentences (proportion or mean example).

Section C — Binomial μ, σ & normal approximation (Read 1 §3)

C1. Binomial mean

ESP test: $n = 25$ trials, $p = 0.2$. What are μ and σ for X = number correct?

C2. Normal approximation conditions

A binomial count has $n = 100$, $p = 0.04$. Can you use the normal approximation with $np \geq 10$ and $n(1-p) \geq 10$? Explain briefly.

C3. Empirical Rule

State the Empirical Rule for a normal distribution (three parts: 68%, 95%, 99%).

C4. Rejection region sketch

Under a normal null curve for a one-proportion test, where is the rejection region for a two-sided test at roughly $\alpha = 0.05$ using the Empirical Rule? How does shaded tail area relate to p-value?

C5. Binomial SD

ESP test: $n = 25$, $p = 0.2$. Compute $\sigma = \sqrt{np(1-p)}$ for X = number correct.

C6. Exact vs normal — small counts

$n = 16$, test $H_0 : p = 0.5$; under H_0, $np = 8$ and $n(1-p) = 8$. Should you rely on the normal approximation?

Section D — Wald 95% CI for p by hand (Read 1 §3)

D1. Wald proportion conditions

For Wald 95% CI $\hat{p} \pm 2\sqrt{\hat{p}(1-\hat{p})/n}$, name sampling, BINS, and large-count requirements.

D2. St. Joe's Wald CI

St. Joe's: $n = 200$ patients, $x = 14$ deaths. Compute $\hat{p}$, SE≈ 0.018, and approximate 95% Wald CI using multiplier 2.

D3. Wald CI — compute endpoints

SRS $n = 220$, $x = 80$ with a part-time job. $\hat{p} = 80/220 \approx 0.364$, SE ≈ 0.032. Compute approximate 95% CI using 2 $\times$ SE.

D4. Exact when Wald fails

$n = 18$, $x = 2$ successes; $n\hat{p} < 10$. Which CI approach is safer than Wald?

D5. Plus Four trigger

St. Joe's small sample: $n = 16$, $x = 3$ deaths. Why might you use Plus Four instead of Wald?

Section E — Standardized z for one proportion (Read 1 §4)

E1. Coin test hypotheses

16 heads in 20 flips; test if coin is fair (two-sided). State H_0, H_a, and define p in words.

E2. Role of p_0

In $z = (\hat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$, what is p_0?

E3. Compute z

16 heads in 20 flips; $H_0 : p = 0.5$. Compute $z = (\hat{p} - 0.5)/\sqrt{0.5(0.5)/20}$ (decimal). Is $|z|$ beyond 2?

E4. Descriptive vs inferential (proportion)

One binary variable from a sample: give one descriptive summary and one inferential tool for estimation.

Section F — Inferential statistics & sampling distribution

F1. Inferential statistics

Define inferential statistics in one or two sentences and contrast with descriptive statistics.

F2. Sampling distribution

Define the sampling distribution of $\hat{p}$ in one sentence.

Section G — Type I/II vocabulary

G1. Significance level

Define the significance level α in hypothesis testing.

G2. Type I error

Define a Type I error in hypothesis testing.

G3. Type II error

Define a Type II error in hypothesis testing.
